Q1.

The number of printers, V, bought during one day from the *Verigood* store can be modelled by a Poisson distribution with mean 4.5

The number of printers, W, bought during one day from the *Winnerprint* store can be modelled by a Poisson distribution with mean 5.5

(a) Find the probability that the total number of printers bought during one day from *Verigood* and *Winnerprint* stores is greater than 10.

(2)

(b) State the circumstance under which the distributional model you used in part **(a)** would not be valid.

(1)

(Total 3 marks)

Q2.

A large office block is busy during the five weekdays, Monday to Friday, and less busy during the two weekend days, Saturday and Sunday. The block is illuminated by fluorescent light tubes which frequently fail and must be replaced with new tubes by John, the caretaker.

The number of fluorescent tubes that fail on a particular weekday can be modelled by a Poisson distribution with mean 1.5.

The number of fluorescent tubes that fail on a particular weekend day can be modelled by a Poisson distribution with mean 0.5.

(a) Find the probability that:

(i) on one particular Monday, exactly 3 fluorescent light tubes fail;

(2)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(ii) during the two days of a weekend, more than 1 fluorescent light tube fails;

(2)

(iii) during a complete seven-day week, fewer than 10 fluorescent light tubes fail.

(4)

(b) John keeps a supply of new fluorescent light tubes. More new tubes are delivered every Monday morning to replace those that he has used during the previous week. John wants the probability that he runs out of new tubes before the next Monday morning to be less than 1 per cent. Find the minimum number of new tubes that he should have available on a Monday morning.

(2)

(c) Give a reason why a Poisson distribution with mean 0.375 is unlikely to provide a satisfactory model for the number of fluorescent light tubes that fail between 1 am and 7 am on a weekday.

(1)

(Total 11 marks)

For more help, please visit our website www.exampaperspractice.co.uk

Q3.

Gamma-ray bursts (GRBs) are pulses of gamma rays lasting a few seconds, which are produced by explosions in distant galaxies. They are detected by satellites in orbit around Earth. One particular satellite detects GRBs at a constant average rate of 3.5 per week (7 days).

You may assume that the detection of GRBs by this satellite may be modelled by a Poisson distribution.

(a) Find the probability that the satellite detects:

(i) exactly 4 GRBs during one particular week;

(2)

(ii) at least 2 GRBs on one particular **day**;

(3)

(iii) more than 10 GRBs but fewer than 20 GRBs during the 28 days of February 2013.

(3)

(b) Give **one** reason, apart from the constant average rate, why it is likely that the detection of GRBs by this satellite may be modelled by a Poisson distribution.

(1)

(Total 9 marks)

Q4.

The demand for a *WWSatNav* at a superstore may be modelled by a Poisson distribution with a mean of 2.5 per day. The superstore is open 6 days each week, from Monday morning to Saturday evening.

(a) Determine the probability that the demand for *WWSatNavs* during a particular week is at most 18.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(2)

(b) The superstore receives a delivery of *WWSatNavs* on each Sunday evening. The manager, Meena, requires that the probability of *WWSatNavs* being out of stock during a week should be at most 5%.

Determine the minimum number of *WWSatNavs* that Meena requires to be in stock after a delivery.

(2)

(Total 4 marks)

Q5.

(a) A discrete random variable X has a probability function defined by

$$P(X = x) = \frac{e^{-\lambda}\lambda^x}{x!} \quad \text{for } x = 0,\ 1,\ 2,\ 3,\ 4,\ \_\_\_$$

(i) State the name of the distribution of X.

(1)

For more help, please visit our website www.exampaperspractice.co.uk

(ii) Write down, in terms of λ, expressions for $E(3X - 1)$ and $Var(3X - 1)$.

(2)

(iii) Write down an expression for $P(X = x + 1)$, and hence show that

$$P(X = x + 1) = \frac{\lambda}{x+1} P(X = x)$$

(3)

(b) The number of cars and the number of coaches passing a certain road junction may be modelled by independent Poisson distributions.

(i) On a winter morning, an average of 500 cars per hour and an average of 10 coaches per hour pass this junction.

Determine the probability that a total of at least 10 such vehicles pass this junction during a particular 1-minute interval on a winter morning.

(3)

(ii) On a summer morning, an average of 836 cars per hour and an average of 22 coaches per hour pass this junction.

Calculate the probability that a total of at most 3 such vehicles pass this junction during a particular 1-minute interval on a summer morning. Give your answer to two significant figures.

(3)

(Total 12 marks)

Q6.

(a) The number of **minor** accidents occurring each year at RapidNut engineering company may be modelled by the random variable X having a Poisson distribution with mean 8.5.

EXAM PAPERS PRACTICE

Determine the probability that, in any particular year, there are:

©2025 Exam Papers Practice. All rights reserved.

(i) at least 9 minor accidents;

(2)

(ii) more than 5 but fewer than 10 minor accidents.

(3)

(b) The number of **major** accidents occurring each year at RapidNut engineering company may be modelled by the random variable Y having a Poisson distribution with mean 1.5.

Calculate the probability that, in any particular year, there are fewer than 2 major accidents.

(2)

(c) The **total** number of minor and major accidents occurring each year at RapidNut engineering company may be modelled by the random variable T having the probability distribution

For more help, please visit our website www.exampaperspractice.co.uk

$$P(T = t) = \begin{cases} \dfrac{e^{-\lambda}\lambda^t}{t!} & t = 0, 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

Assuming that the number of minor accidents is independent of the number of major accidents:

(i) state the value of λ;

(1)

(ii) determine $P(T > 16)$;

(2)

(iii) calculate the probability that there will be a total of more than 16 accidents in each of at least two out of three years, giving your answer to four decimal places.

(3)

(Total 13 marks)

Q7.

Lucy is the captain of her school's cricket team.

The number of catches, X, taken by Lucy during any particular cricket match may be modelled by a Poisson distribution with mean 0.6.

The number of run-outs, Y, effected by Lucy during any particular cricket match may be modelled by a Poisson distribution with mean 0.15.

(a) Find:

(i) $P(X \leq 1)$;

(1)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(ii) $P(X \leq 1$ **and** $Y \geq 1)$.

(4)

(b) State the assumption that you made in answering part (a)(ii).

(1)

(c) During a particular season, Lucy plays in 16 cricket matches.

(i) Calculate the probability that the number of catches taken by Lucy during this season is exactly 10.

(2)

(ii) Determine the probability that the total number of catches taken and run-outs effected by Lucy during this season is at least 15.

(3)

(Total 11 marks)

Q8.

For more help, please visit our website www.exampaperspractice.co.uk

The number of cars passing a speed camera on a main road between 9.30 am and 11.30 am may be modelled by a Poisson distribution with a mean rate of 2.6 per minute.

(a) (i) Write down the distribution of X, the number of cars passing the speed camera during a 5-minute interval between 9.30 am and 11.30 am.

(1)

(ii) Determine P(X = 20).

(2)

(iii) Determine P(6 ≤ X ≤ 18).

(3)

(b) Give **two** reasons why a Poisson distribution with mean 2.6 may not be a suitable model for the number of cars passing the speed camera during a 1-minute interval between 8.00 am and 9.30 am on weekdays.

(2)

(c) When n cars pass the speed camera, the number of cars, Y, that exceed 60 mph may be modelled by the distribution B(n, 0:2).

Given that n = 20, determine P(Y ≥ 5).

(2)

(d) Stating a necessary assumption, calculate the probability that, during a given 5-minute interval between 9.30 am and 11.30 am, exactly 20 cars pass the speed camera of which at least 5 are exceeding 60 mph.

(3)
(Total 13 marks)

Q9.

The random variable Y has a Poisson distribution with E(Y) = 2.5.

Given that Z = 4Y + 30:

(a) show that Var(Z) = E(Z);

(3)

(b) give a reason why the distribution of Z is **not** Poisson.

(1)
(Total 4 marks)

Q10.

(a) In a remote African village, it is known that 70 per cent of the villagers have a particular blood disorder. A medical research student selects 25 of the villagers at random.

Using a binomial distribution, calculate the probability that more than 15 of these 25 villagers have this blood disorder.

(3)

(b) (i) In towns and cities in Asia, the number of people who have this blood

EXAM PAPERS PRACTICE
© 2025 Exam Papers Practice. All rights reserved.

For more help, please visit our website www.exampaperspractice.co.uk

disorder may be modelled by a Poisson distribution with a mean of 2.6 per 100 000 people.

A town in Asia with a population of 100 000 is selected. Determine the probability that at most 5 people have this blood disorder.

(1)

(ii) In towns and cities in South America, the number of people who have this blood disorder may be modelled by a Poisson distribution with a mean of 49 per **million** people.

A town in South America with a population of 100 000 is selected. Calculate the probability that exactly 10 people have this blood disorder.

(3)

(iii) The random variable T denotes the **total** number of people in the two selected towns who have this blood disorder.

Write down the distribution of T and hence determine P(T > 16).

(3)
(Total 10 marks)

Q11.

The number of telephone calls received, during an 8-hour period, by an IT company that request an urgent visit by an engineer may be modelled by a Poisson distribution with a mean of 7.

(a) Determine the probability that, during a given 8-hour period, the number of telephone calls received that request an urgent visit by an engineer is:

(i) at most 5;

(1)

(ii) exactly 7;

(2)

(iii) at least 5 but fewer than 10.

(3)

(b) Write down the distribution for the number of telephone calls received each hour that request an urgent visit by an engineer.

(1)

(c) The IT company has 4 engineers available for urgent visits and it may be assumed that each of these engineers takes exactly 1 hour for each such visit.

At 10 am on a particular day, all 4 engineers are available for urgent visits.

(i) State the maximum possible number of telephone calls received between 10 am and 11 am that request an urgent visit and for which an engineer is immediately available.

(1)

(ii) Calculate the probability that at 11 am an engineer will **not** be immediately available to make an urgent visit.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

For more help, please visit our website www.exampaperspractice.co.uk

(4)

(d) Give a reason why a Poisson distribution may not be a suitable model for the number of telephone calls per hour received by the IT company that request an urgent visit by an engineer.

(1)

(Total 13 marks)

Q12.

Joe owns two garages, Acefit and Bestjob, each specialising in the fitting of the latest satellite navigation device.

The daily demand, X, for the device at Acefit garage may be modelled by a Poisson distribution with mean 3.6.

The daily demand, Y, for the device at Bestjob garage may be modelled by a Poisson distribution with mean 4.4.

(a) Calculate:

(i) $P(X \leq 3)$;

(1)

(ii) $P(Y = 5)$.

(2)

(b) The total daily demand for the device at Joe's two garages is denoted by T.

(i) Write down the distribution of T, stating any assumption that you make.

(2)

EXAM PAPERS PRACTICE

(ii) Determine $P(6 < T < 12)$.

(3)

©2025 Exam Papers Practice. All rights reserved

(iii) Calculate the probability that the total demand for the device will exceed 14 on each of two consecutive days. Give your answer to one significant figure.

(4)

(iv) Determine the minimum number of devices that Joe should have in stock if he is to meet his total demand on at least 99% of days.

(2)

(Total 14 marks)

Q13.

John works from home. The number of business letters, X, that he receives on a weekday may be modelled by a Poisson distribution with mean 5.0.

The number of private letters, Y, that he receives on a weekday may be modelled by a Poisson distribution with mean 1.5.

(a) Find, for a given weekday:

For more help, please visit our website www.exampaperspractice.co.uk

(i) P(X < 4);

(2)

(ii) P(Y = 4).

(2)

(b) (i) Assuming that X and Y are independent random variables, determine the probability that, on a given weekday, John receives a **total** of more than 5 business and private letters.

(3)

(ii) Hence calculate the probability that John receives a **total** of more than 5 business and private letters on at least 7 out of 8 given weekdays.

(3)

(c) The numbers of letters received by John's neighbour, Brenda, on 10 consecutive weekdays are

15 8 14 7 6 8 2 8 9 3

(i) Calculate the mean and the variance of these data.

(2)

(ii) State, giving a reason based on your answers to part (c)(i), whether or not a Poisson distribution might provide a suitable model for the number of letters received by Brenda on a weekday.

(2)

(Total 14 marks)

Q14.

A new information technology centre is advertising places on its one-week residential computer courses.

EXAM PAPERS PRACTICE

(a) The number of places, X, booked each week on the publishing course may be modelled by a Poisson distribution with a mean of 9.0.

©2025 Exam Papers Practice. All rights reserved.

(i) State the standard deviation of X.

(1)

(ii) Calculate P(6 < X < 12).

(3)

(b) The number of places booked each week on the web design course may be modelled by a Poisson distribution with a mean of 2.5.

(i) Write down the distribution for T, the **total** number of places booked each week on the publishing and web design courses.

(1)

(ii) Hence calculate the probability that, during a given week, a total of fewer than 2 places are booked.

(3)

(c) The number of places booked on the database course during each of a random

For more help, please visit our website www.exampaperspractice.co.uk

sample of 10 weeks is as follows:

14 15 8 16 18 4 10 12 15 8

By calculating appropriate numerical measures, state, with a reason, whether or not the Poisson distribution Po(12.0) could provide a suitable model for the number of places booked each week on the database course.

(3)

(Total 11 marks)

Q15.

(a) The number of genuine telephone calls, X, made to the emergency services each hour may be modelled by a Poisson distribution with mean 6.5.

Determine the probability that, in a given hour, there will be fewer than 6 genuine telephone calls made to the emergency services.

(2)

(b) The number of bogus telephone calls, Y, made to the emergency services each hour may be modelled by a Poisson distribution with mean 1.5.

Calculate the probability that, in a given hour, there will be at least 1 bogus telephone call made to the emergency services.

(3)

(c) Assuming that X and Y are independent Poisson variables, find:

(i) $\text{P}(X + Y = 6)$;

(3)

(ii) $\text{P}(Y \geq 1 \mid X < 6)$.

(1)

(Total 9 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q16.

(a) The number of telephone calls, X, received per hour for Dr Able may be modelled by a Poisson distribution with mean 6.

Determine $\text{P}(X = 8)$.

(2)

(b) The number of telephone calls, Y, received per hour for Dr Bracken may be modelled by a Poisson distribution with mean λ and standard deviation 3.

(i) Write down the value of λ.

(1)

(ii) Determine $\text{P}(Y > \lambda)$.

(2)

(c) (i) Assuming that X and Y are independent Poisson variables, write down the distribution of the **total** number of telephone calls received per hour for Dr

For more help, please visit our website www.exampaperspractice.co.uk

Able and Dr Bracken.

(1)

(ii) Determine the probability that a total of at most 20 telephone calls will be received during any one-hour period.

(1)

(iii) The total number of telephone calls received during each of 6 one-hour periods is to be recorded. Calculate the probability that a total of at least 21 telephone calls will be received during exactly 4 of these one-hour periods.

(3)

(Total 10 marks)

Q17.

The number of accidents, X, occurring during one week at Joanne's place of work can be modelled by a Poisson distribution with a mean of 0.7.

The number of accidents, Y, occurring during one week at Pete's place of work can be modelled by a Poisson distribution with a mean of 1.3.

(a) (i) Determine $P(X < 3)$.

(1)

(ii) Calculate $P(Y = 2)$.

(2)

(b) Find the probability that, during a particular week, there are at least 4 accidents in total at these two places of work.

(3)

(Total 6 marks)

EXAM PAPERS PRACTICE

Q18.

©2025 Exam Papers Practice. All rights reserved.

The number of computers, A, bought during one day from the Amplebuy computer store can be modelled by a Poisson distribution with a mean of 3.5.

The number of computers, B, bought during one day from the Bestbuy computer store can be modelled by a Poisson distribution with a mean of 5.0.

(a) (i) Calculate $P(A = 4)$.

(2)

(ii) Determine $P(B \leq 6)$.

(1)

(iii) Find the probability that a total of fewer than 10 computers is bought from these two stores on one particular day.

(3)

(b) Calculate the probability that a total of fewer than 10 computers is bought from these two stores on at least 4 out of 5 consecutive days.

(3)

For more help, please visit our website www.exampaperspractice.co.uk

(c) The numbers of computers bought from the Choicebuy computer store over a 10-day period are recorded as

8 12 6 6 9 15 10 8 6 12

(i) Calculate the mean and variance of these data.

(2)

(ii) State, giving a reason based on your results in part (c) (i), whether or not a Poisson distribution provides a suitable model for these data.

(2)

(Total 13 marks)

Q19.

The number of telephone calls per day, X, received by Candice may be modelled by a Poisson distribution with mean 3.5.

The number of e-mails per day, Y, received by Candice may be modelled by a Poisson distribution with mean 6.0.

(a) For any particular day, find:

(i) $P(X = 3)$;

(2)

(ii) $P(Y \geq 5)$

(2)

(b) (i) Write down the distribution of T, the total number of telephone calls and e-mails per day received by Candice.

(1)

(ii) Determine $P(7 \leq T \leq 10)$.

(3)

(iii) Hence calculate the probability that, on each of three consecutive days, Candice will receive a total of at least 7 but at most 10 telephone calls and e-mails.

(2)

(Total 10 marks)

Q20.

The administration department at Hitech College uses two photocopiers, A and B, which operate independently.

The number of breakdowns each week, X, for photocopier A can be modelled by a Poisson distribution with a mean of 0.5.

The number of breakdowns each week, Y, for photocopier B can be modelled by a Poisson distribution with a mean of 2.5.

(a) Calculate:

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

For more help, please visit our website www.exampaperspractice.co.uk

(i) P($X \le 1$);

(2)

(ii) P($Y = 2$).

(2)

(b) (i) In a given week, show that the probability of a total of exactly 2 photocopier breakdowns is 0.224, correct to three decimal places.

(3)

(ii) Hence find the probability that there will be exactly 2 photocopier breakdowns in each of four consecutive weeks.

(2)

(c) The total number of photocopier breakdowns, T, in a 4-week period can be modelled by a Poisson distribution with mean μ.

(i) Find the value of μ.

(1)

(ii) Hence calculate the probability that, in a given 4-week period, there will be a total of at least 18 photocopier breakdowns.

(2)
(Total 12 marks)

Q21.

A study undertaken by Goodhealth Hospital found that the number of patients each month, X, contracting a particular superbug can be modelled by a Poisson distribution with a mean of 1.5.

(a) (i) Calculate P($X = 2$).

EXAM PAPERS PRACTICE

(2)

© 2025 Exam Papers Practice. All rights reserved

(ii) Hence determine the probability that exactly 2 patients will contract this superbug in each of three consecutive months.

(2)

(b) (i) Write down the distribution of Y, the number of patients contracting this superbug in a given 6-month period.

(1)

(ii) Find the probability that at least 12 patients will contract this superbug during a given 6-month period.

(2)

(c) State **two** assumptions implied by the use of a Poisson model for the number of patients contracting this superbug.

(2)
(Total 9 marks)

Q22.

The number of A-grades, X, achieved in total by students at Lowkey School in their

For more help, please visit our website www.exampaperspractice.co.uk

Mathematics examinations each year can be modelled by a Poisson distribution with a mean of 3.

(a) Determine the probability that, during a 5-year period, students at Lowkey School achieve a total of more than 18 A-grades in their Mathematics examinations.

(3)

(b) The number of A-grades, Y, achieved in total by students at Lowkey School in their English examinations each year can be modelled by a Poisson distribution with a mean of 7.

(i) Determine the probability that, during a year, students at Lowkey School achieve a total of fewer than 15 A-grades in their Mathematics and English examinations.

(3)

(ii) What assumption did you make in answering part (b)(i)?

(1)

(Total 7 marks)

Q23.

The number of letters, Y, delivered each day to Martin's house can be modelled by a Poisson distribution with mean 1.9. Assume that the number of letters delivered on any day is independent of the number of letters delivered on any other day.

(a) (i) Calculate the value of P(Y = 2).

(2)

(ii) Hence determine the probability that exactly 2 letters will be delivered on each of five consecutive days.

(2)

EXAM PAPERS PRACTICE

(b) (i) Write down the distribution of X, the number of letters delivered during a 5-day period.

©2025 Exam Papers Practice. All rights reserved.

(1)

(ii) Find the probability that at least 10 letters will be delivered during a 5-day period.

(2)

(iii) Hence calculate the probability that at least 10 letters will be delivered during exactly three out of five consecutive 5-day periods.

(2)

(Total 9 marks)

Q24.

The number of cars, X, passing along a road each minute can be modelled by a Poisson distribution with a mean of 2.6.

(a) Calculate P(X = 2).

(2)

For more help, please visit our website www.exampaperspractice.co.uk

(b) (i) Write down the distribution of Y, the number of cars passing along this road in a 5-minute interval.

(1)

(ii) Hence calculate the probability that at least 15 cars pass along this road in each of four successive 5-minute intervals.

(4)

(Total 7 marks)

Q25.

The number of mobile phone calls, X, that Pat makes each day to her boyfriend James has a Poisson distribution with a mean of 4.0. The number of text messages, Y, that James sends each day to Pat has a Poisson distribution with a mean of 3.5. Assume that X and Y are independent random variables.

(a) Calculate the probability that, on a randomly chosen day:

(i) Pat phones James on more than 8 occasions;

(2)

(ii) James sends fewer than 2 text messages to Pat.

(2)

(b) The total number of phone calls made and text messages sent, T, is a Poisson random variable with mean λ.

(i) Write down the value of λ.

(1)

(ii) Calculate the probability that, on a randomly selected day, T is at least eleven.

(2)

(Total 7 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q26.

The number of strikes per game, obtained by a tenpin bowler, can be modelled by a Poisson distribution with mean λ. For the first game played, λ = 0.2 and, for each subsequent game played, λ = 1.1.

In a match consisting of three consecutive independent games, calculate the probability that the tenpin bowler will obtain a total of between 3 and 5 strikes, inclusive.

(Total 5 marks)

Q27.

The number of vehicles arriving at a toll bridge during a 5-minute period can be modelled by a Poisson distribution with mean 3.6.

(a) Calculate,

(i) the probability that at least 3 vehicles arrive in a 5-minute period,

(3)

For more help, please visit our website www.exampaperspractice.co.uk

(ii) the probability that at least 3 vehicles arrive in each of three successive 5-minute periods.

(3)

(b) Show that the probability that no vehicles arrive in a 10-minute period is 0.0007, correct to four decimal places.

(3)
(Total 9 marks)

Q28.

At a petrol station the number of 'drive-outs' (motorists who obtain petrol and then drive away without paying) may be modelled by a Poisson distribution with mean 0.8 per day.

(a) Find the probability that, on a particular day, the number of 'drive-outs' is:

(i) 2 or fewer;

(ii) more than 1 but fewer than 3.

(4)

(b) (i) Evaluate the mean and the standard deviation of the number of 'drive-outs' during a 5-day period.

(3)

(ii) Find the probability that there are fewer than 3 'drive-outs' during a 5-day period.

(1)
(Total 8 marks)

Q29.

Messages by email are received by Tariq's personal computer, independently, at random, at an average rate of 2.4 per day.

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

(a) Name a distribution which provides a suitable model for the number of email messages received per day by Tariq's personal computer.

(1)

(b) Find the probability that the number of messages received on a particular day is:

(i) two or fewer;

(ii) exactly four.

(3)

(c) Find the probability that the number of messages received during a five-day period is:

(i) fewer than six;

(ii) more than 17.

(5)
(Total 9 marks)

For more help, please visit our website www.exampaperspractice.co.uk

Q30.

Karen is an engineer who is responsible for maintaining a textile machine during the night shift. She is called when the machine operator believes that the machine needs adjustment or repair.

(a) The number of times Karen is called during a night shift may be modelled by a Poisson distribution with mean 0.8. Find the probability that, on a particular shift, she is called:

(i) two or fewer times;

(1)

(ii) at least once;

(2)

(iii) exactly once.

(2)

(b) Karen is now made responsible for 5 machines. These machines are operated independently. For each machine, the number of times she is called during a night shift may be modelled by a Poisson distribution with mean 0.8.

(i) State the distribution of the total number of times she is called during a night shift.

(1)

(ii) Find the probability that, during a night shift, she is called 5 or more times.

(2)

(iii) Find the probability that, during a night shift, she is called at least once to each of the 5 machines.

(3)

(Total 11 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q31.

The number of family groups entering a museum on an August afternoon may be modelled by a Poisson distribution with mean 0.60 per minute.

(a) Find the probability that during a particular minute 2 or fewer such groups enter the museum.

(2)

(b) Find the probability that a particular **five-minute** interval more than 4 such groups enter the museum.

(3)

(c) Explain why a Poisson distribution would not provide a suitable model for the number of **visitors** entering the museum during one-minute intervals on an August afternoon.

(2)

(Total 7 marks)

Q32.

For more help, please visit our website www.exampaperspractice.co.uk

The number of cyclists passing a point on a cycle track alongside a main road into Oxford, during the morning rush hour, may be modelled by a Poisson distribution with mean 2.8 per minute.

(a) Find the probability that the number of cyclists passing the point during a particular minute in the morning rush hour is:

(i) one or fewer;

(ii) 3 or more.

(3)

(b) Find the probability that fewer than ten cyclists pass the point during a **five** minute interval in the morning rush hour.

(3)

(c) The numbers of cyclists passing the point during ten, randomly selected, one minute intervals over a twenty four hour period, were:

0 0 5 1 0 4 1 0 0 2

(i) Calculate the mean and variance of this sample.

(2)

(ii) Give a reason, based on your results in part (c)(i), why the Poisson distribution is unlikely to provide an adequate model for the number of cyclists passing the point over a twenty four hour period.

(1)

(iii) Give a reason, **not** based on your results in part (c)(i), why the Poisson distribution is unlikely to provide an adequate model for the number of cyclists passing the point over a twenty four hour period.

(1)

(Total 10 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

Q33.

Bronwen runs a post office in a large village. The number of registered letters posted at this office may be modelled by a Poisson distribution with mean 1.4 per day.

(a) Find the probability that at this post office:

(i) two of fewer registered letters are posted on a particular day;

(ii) a total of four or more registered letters are posted on two consecutive days.

(5)

(b) The village also contains a post office run by Gopal. Here the number of registered letters posted may be modelled by a Poisson distribution with mean 2.4 per day.

Find the probability that, on a particular day, the number of registered letters posted at Gopal's post office is less than 4.

(2)

(c) The numbers of registered letters posted at the two post offices are independent. Find the probability that, on a particular day, the total number of registered letters

For more help, please visit our website www.exampaperspractice.co.uk

posted at the two post offices is more than six.

(3)

(d) Give one reason why the Poisson distribution might **not** provide a suitable model for the number of registered letters posted daily at a post office.

(1)
(Total 11 marks)

EXAM PAPERS PRACTICE
©2025 Exam Papers Practice. All rights reserved.

For more help, please visit our website www.exampaperspractice.co.uk